UNIVERSITY OF ABERDEEN

CELEBRATING 525 YEARS 1495 – 2020

ABERDEEN 2040

Revision – Parts 1 & 2

Data Mining & Visualisation

Today...

- Exam-style questions that cover some of the content from Parts 1 & 2 of the course.

Decision Trees

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Decision Trees

Consider the following dataset, which shows a list of potential customers, and whether or not they purchased a particular product. You have been asked to construct a decision tree that can be used to predict whether someone is likely to have **Purchased** the product or not.

Use the 'Useful Log2 Calculations' table below to help you.

ID	Age	Salary	Purchased
1	18 – 29	30k – 39k	Yes
2	18 – 29	20k – 29k	No
3	18 – 29	20k – 29k	No
4	18 – 29	30k – 39k	Yes
5	30 – 39	30k – 39k	Yes
6	30 – 39	30k – 39k	Yes
7	30 – 39	20k – 29k	No
8	30 – 39	20k – 29k	No

Useful Log_2 Calculations		
$log_2(0) = -\infty$ (treat as 0)	$log_2(0.5) = -1$	$log_2(1) = 0$

Decision Trees

(a) The entropy of the full dataset (before splitting on any variables) is 1. Prove this by providing the calculation which shows that this is the case. [2 mark]

(b) Compute the Information Gain for the ‘Age’ attribute selected as the root node. Make sure to show your calculations. [6 mark]

(c) Compute the Information Gain for the ‘Salary’ attribute selected as the root node. Make sure to show your calculations. [6 mark]

(d) Based on your answers for part (b) and (c), which attribute would you split on at the root of the tree; Age Range or Salary? [1 mark]

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Decision Trees

(b) Compute the Information Gain for the ‘Age’ attribute selected as the root node. Make sure to show your calculations. [6 mark]

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Decision Trees

(c) Compute the Information Gain for the 'Salary' attribute selected as the root node. Make sure to show your calculations. [6 mark]

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Decision Trees

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Naïve Bayes Classification

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Naïve Bayes

Consider the following table, which shows some of the words included within a set of emails (vocabulary list).

Each email is classified as either **spam** (**S**), or **not spam** (**N**).

Email ID	Email Vocabulary List	Class
1	hello friend	N
2	greetings bank candidate	N
3	greetings money bank friend	S
4	greetings friend	N
5	hello job friend	N
6	Hello candidate	N
7	greetings millionaire money candidate	S
8	hello bank job friend	N

Naïve Bayes

Based on the data given in the above table:

(a) Calculate the prior probability of a document occurring in each class, i.e. P(S) and P(N) [2 mark]

(b) For each word in the email vocabulary list, calculate the conditional word probability given a class label, i.e. P(money | S), P(money | N), etc.

Note that smoothing methods should not be used. [8 mark]

(c) Using the information you have already computed from (b), use Naïve Bayes classification to state the predicted label of a new email with the vocabulary list: {greetings bank}.

Note that you can use fractions to make your calculations easier. [5 mark]

Naïve Bayes

(a) Calculate the prior probability of a document occurring in each class, i.e. P(S) and P(N) [2 mark]

Email ID	Email Vocabulary List	Class
1	hello friend	N
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3	greetings money bank friend	S
4	greetings friend	N
5	hello job friend	N
6	Hello candidate	N
7	greetings millionaire money candidate	S
8	hello bank job friend	N

Naïve Bayes

(b) For each word in the email vocabulary list, calculate the conditional word probability given a class label, i.e. P(money | S), P(money | N), etc. Note that smoothing methods should not be used. [8 mark]

Email ID	Email Vocabulary List	Class
1	hello friend	N
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3	greetings money bank friend	S
4	greetings friend	N
5	hello job friend	N
6	Hello candidate	N
7	greetings millionaire money candidate	S
8	hello bank job friend	N

Naïve Bayes

(c) Using the information you have already computed from (b), use Naïve Bayes classification to state the predicted label of a new email with the vocabulary list: {greetings bank}. Note that you can use fractions to make your calculations easier. [5 mark]

Email ID	Email Vocabulary List	Class
1	hello friend	N
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7	greetings millionaire money candidate	S
8	hello bank job friend	N

MCQ Example Questions

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MCQ Question 1

1. Which level of measurement would you use to classify temperature ({Cold; Warm; Hot})?

(a) Nominal
(b) Ordinal
(c) Interval
(d) Ratio

MCQ Question 2

2. Which of the following is an example of nominal data?

(a) Temperature: 22◦C
(b) Education level: {High School; Bachelor's Degree; Master's Degree; PhD}
(c) Nationality: {British; Chinese; French; Italian}
(d) Age: 21

MCQ Question 3

3. Which of the following is a valid example of data upsampling in binary classification?

(a) Copying across the value held within a similar row to a row of missing data
(b) Duplicate instances of the minority class such that its n = n of the majority class
(c) Remove instances of the majority class such that its n = n of the minority class
(d) Converting categorical variables to numerical (Boolean) values

MCQ Question 4

4. Consider the following dataset transformation (Table 1):

This use of converting a continuous numerical variable into a discrete categorical variable is an example of:

(a) Dummy coding
(b) Data downsampling
(c) Split-apply-combine
(d) Bucketing

city	population
Aberdeen	199,000
Beijing	22,189,000
London	8,866,000
Monaco	39,000

→

city	population
Aberdeen	medium
Beijing	high
London	high
Monaco	low

Table 1: A dataset containing the population of cities.

MCQ Question 5

5. Which of the below answers is not one of the layered core components of data visualisations, as outlined in the Grammar of Graphics?

(a) Data
(b) Correlation Projections
(c) Statistical Transformations
(d) Geometric Objects

MCQ Question 6

6. What does the following formula represent: $r_{xy} = \frac{\text{cov}(x,y)}{s_X s_Y}$

(a) Sample Pearson correlation coefficient
(b) R2
(c) Population Pearson correlation coefficient
(d) Pseudo-R2 for classification

MCQ Question 7

7. Which answer best describes the relationship between the two variables shown in Figure 1?

(a) A strong positive correlation
(b) A strong negative correlation
(c) A weak positive correlation
(d) A weak negative correlation

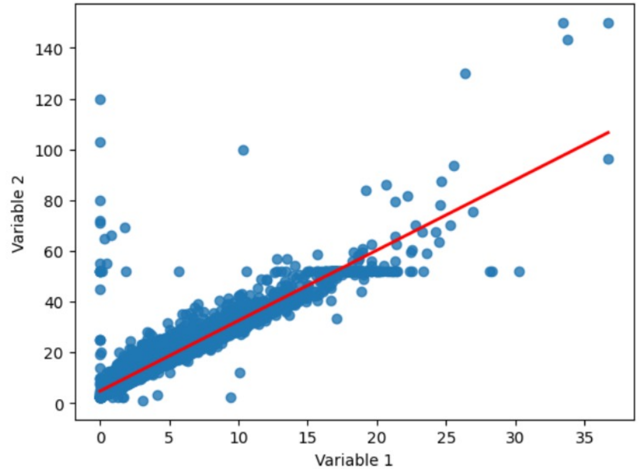

Figure 1: Two variables plotted on a scatter plot.

MCQ Question 8

8. The letters in the abbreviation ‘NHST’ stand for?

(a) Null Hypothesis Statistical Testing
(b) Normal Hypothesis Sample Testing
(c) Nonparametric Hypothesis Statistical Technique
(d) Numerical Hypothesis Sampling Technique

MCQ Question 9

9. The outcome of a NHST will depend on which three inter-related factors?

(a) The correlation coefficient, p-value, and sample size
(b) The sample size, correlation coefficient, and the mean
(c) The confidence interval, standard error, and p-value
(d) The p-value, sample size, and effect size

MCQ Question 10

10. Which metric for evaluating a regression model has the following equation: $\sqrt{\frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$

(a) SSE
(b) MSE
(c) RMSE
(d) R2

MCQ Question 11

11. For the regression model: $\hat{y} = x_1 + x_2$, which variable(s) are considered the ‘independent variable(s)’ (IV)?

(a) $\hat{y}$
(b) x_1 only
(c) x_2 only
(d) x_1 and x_2

MCQ Question 12

12. The entropy of a (fair) coin flip is:

(a) 0
(b) 0.5
(c) 0.75
(d) 1

MCQ Question 13

13. Which of the following statements about pruning a decision tree is untrue?

(a) Pre-pruning involves building the complete tree, then replacing some non-leaf nodes with leaf nodes
(b) Pre-pruning involves stopping the tree-building algorithm before it fully classifies the data
(c) Pruning can prevent the model from overfitting on the training data
(d) Pruning is a regularization method that reduces the complexity of the final model

MCQ Question 14

14. Which of the following methods are not considered to be a discriminative classifier ?

(a) Logistic regression
(b) Naïve Bayes classification
(c) Decision trees
(d) Support vector machines

MCQ Question 15

15. What is the role of the kernel function in support vector machines?

(a) To reduce the dimensionality of the input data
(b) To balance the classes for linear separation
(c) To remove data points which make linear separation impossible
(d) To map input data into higher-dimensional space for linear separation